International Baccalaureate Diploma Program

Environmental Systems & Societies Extended Essay 2023

Topic: Comparative effects of enhanced UV infiltrations due to depletion of ozone on plants

Research question: How far is the enhanced infiltration of UVA radiations (315-400nm) due to ozone depletion less detrimental than UVB radiations(280-315nm) for the germination and growth of Gerbera (*Gerbera L.*) and Chamomile (*Matricaria chamomilla*) flowering plants?

Personal code:

Word count: 3987

Table of Contents

1. Introduction: 3

2. Background information: 6

3. Hypothesis: 8

3.1 Null Hypothesis: 8

3.2 Alternate Hypothesis: 8

4. Variables: 9

4.1 Independent Variables: 9

4.2 Dependent Variables: 9

5. Methodology: 9

5.1 Experimental Set-up: 9

5.1.1 Seed preparation: 9

Figure 1: Seed preparation 9

5.1.2 Seed germination and plant growth set-up: 10

5.2 Sampling strategy & justification: 11

5.3 Photographic Evidence: 12

6. Data collection & processing: 13

7. Result Analysis & Discussion: 21

7.1 Qualitative observation: 21

7.2 Quantitative observation: 22

8. Conclusion: 25

9. Limitation and possible improvements: 26

10. Future scope: 27

11. Bibliography: 28

1. Introduction:

As per the **World Health Organization**,[1] Sun is by far the biggest source of UV radiations on Earth. This radiation spectrum can be divided into 3: UV-A (320nm-400nm), UV-B (280nm-320nm) and UV-C (200nm-280nm), but UVB radiations are the riskiest. According to **UCAR Centre for Science Education**,[2] the ozone in the stratospheric layer absorbs most of the UVC and UVB radiations from sunlight as it passes through the atmosphere, but UVA radiations are not absorbed, which is of least concern because they have the longest wavelength out of the three, so are least energetic and harmful. Additionally, while UVC radiations have the same short wavelengths and high energy as UVB radiations, they get absorbed by oxygen molecules that are present in the atmosphere, so do not pose a threat to life on Earth. Thus, eliminating the exposure risks of the rest of the UV radiations except the enhanced infiltration of UVB radiations which has aggravated as the ozone layer is getting thinner, and the protective filter activity of the atmosphere is progressively reducing. The phenomenon has put the scientists and governments of the world on edge as according to 18-month long rigorous studies conducted by **International Ozone Trends Panel**[3], regions at 30^0 and 60^0 latitudes have witnessed a significant thinning by 1.7% and 3.0%, respectively in the ozone layer from the years 1969 to 1986. The concern has been further reiterated by **NASA**[4] scientists, as enhanced depletion of 4.3% has occurred in the present times owing to augmentation of chlorine in the atmosphere emitted from the use of CFCs- Chlorofluorocarbons, HCFCs- Hydrochlorofluorocarbons, and halons, present in ACs, refrigerators, fire extinguishers, etc.[5]

Scope to the investigation was provided through various detailed investigations conducted by academic scholars like **Aditee Potdar** in Journal of **Ecosystem & Ecography**[6] who has confirmed the deleterious impacts of exposure to UVB radiations on plant productivity mainly because of their inescapability. Thus, even the slightest

---

[1] *UV?, What et al. "Radiation: Ultraviolet (UV) Radiation". Who.Int, 2022, https://www.who.int/news-room/questions-and-answers/item/radiation-ultraviolet-(uv). Accessed 23 Aug 2022.*

[2] *"Ultraviolet (UV) Radiation | Center For Science Education". Scied.Ucar.Edu, 2022, https://scied.ucar.edu/learning-zone/atmosphere/ultraviolet-uv-radiation. Accessed 23 Aug 2022.*

[3] *"Potential Impacts Of Increased Solar UV-B On Global Plant Productivity". Ciesin.Columbia.Edu, 2022, http://www.ciesin.columbia.edu/docs/011-467/011-467.html. Accessed 22 Aug 2022.*

[4] *"Potential Impacts Of Increased Solar UV-B On Global Plant Productivity". Ciesin.Columbia.Edu, 2022, http://www.ciesin.columbia.edu/docs/011-467/011-467.html. Accessed 25 Jul 2022.*

[5] *Department of the Environment and Energy. "Department of Agriculture, Water and the Environment." Department of Agriculture, Water and the Environment, 2018, www.environment.gov.au/protection/ozone/publications/ozone-depleting-substances-factsheet#:~:text=Ozone%20depleting%20substances%20are%20chemicals. Accessed 25 Nov. 2022.*

[6] *Potdar, Aditee. "Business Initiatives For Conservation And Enhancement Of Biodiversity". Journal Of Ecosystem & Ecography, vol 01, no. S1, 2012. OMICS Publishing Group, doi:10.4172/2157-7625.s1.009. Accessed 25 Dec 2022.*

enhancement in the infiltration of these radiations can have far reaching impacts on the growth and final yield of the plants. Not only that, but other adverse impacts like inducing of intra and inter-specific competition, decline in photosynthetic rates, increased vulnerability to diseases and modification in the structure and pigmentation of plants have also been observed. Plant specialists like **Lucas Vanhaelewyn et al**[7] have corroborated in their scholastic paper, **"Ultraviolet Radiation…Plant-Microorganism context",** that plants show higher sensitivity to UVB radiations owing to their immobile nature. These radiations are known to cause permanent damage to the cell membranes and other plant organelles like mitochondria, chloroplasts, DNA which hamper their life processes like growth, repair, respiration, circulatory system, etc., consequently, leading to financial losses among farmers.[8] Validation has also come through data shared by the **Climate Policy Watcher**,[9] which has stated that there has been a substantial decline in the global yield due to exposure to enhanced UVB radiations. The direct impact of this has been projected to result in a 20-25% decrease in global crop production, as confirmed by academic scientists like **Fiscus et al**.[10]

Although the detrimental impacts of exposure to enhanced UVB radiations have been reiterated by many scientists, but the findings have faced some contradictions as well through the works of some academic explorers like, **Timothy Zwier**[11] of **Purdue university in West Lafayette, Indiana**.[12] He has found that plants have adapted and developed resistivity towards UVB radiation exposure. As per his findings, these plants have established their internal defense mechanism when facing prolonged exposure to sun while photosynthesizing. They can create a composition of various chemicals known as "Sinapate esters" which provides them the necessary protection against UVB radiations. Another composition is of an aromatic chemical, Sinopyl Malate, which accumulates on top of the leaves and protects the plant.[13] Further research showed that this layer has the

---

[7] *Vanhaelewyn, Lucas et al. "Ultraviolet Radiation From A Plant Perspective: The Plant-Microorganism Context". Frontiers In Plant Science, vol 11, 2020. Frontiers Media SA, doi:10.3389/fpls.2020.597642. Accessed 23 Aug 2022.*

[8] *"Specific Effects Of UVB Radiation On Plants - Ultraviolet Radiation". Climate Policy Watcher, 2020, https://www.climate-policy-watcher.org/ultraviolet-radiation-2/specific-effects-of-uvb-radiation-on-plants.html. Accessed 25 Dec 2022.*

[9] *"Specific Effects Of UVB Radiation On Plants - Ultraviolet Radiation". Climate Policy Watcher, 2020, https://www.climate-policy-watcher.org/ultraviolet-radiation-2/specific-effects-of-uvb-radiation-on-plants.html. Accessed 25 Dec 2022.*

[10] *Fiscus, Edwin L., and Fitzgerald L. Booker. "Is Increased UV-B a Threat to Crop Photosynthesis and Productivity?" Photosynthesis Research, vol. 43, no. 2, Feb. 1995, pp. 81–92, doi:10.1007/bf00042965. Accessed 1 Dec. 2022.*

[11] *"Zwier Lab - Purdue University". Chem.Purdue.Edu, 2022, http://www.chem.purdue.edu/zwier/?_ga=2.97920279.906543378.1613986145-1359163524.1613986145. Accessed 22 Jun 2022.*

[12] *"Study Gets To The Roots Of Plants' Natural Sunscreen - Purdue University". Purdue.Edu, 2022, https://www.purdue.edu/newsroom/releases/2014/Q4/study-gets-to-the-roots-of-plants-natural-sunscreen.html. Accessed 22 Jun 2022.*

[13] *Nguyen, V. P. Thinh et al. "Sinapic Acid And Sinapate Esters In Brassica: Innate Accumulation, Biosynthesis, Accessibility Via Chemical Synthesis Or Recovery From Biomass, And Biological Activities". Frontiers In Chemistry, vol 9, 2021. Frontiers Media SA, doi:10.3389/fchem.2021.664602. Accessed 8 Feb 2023.*

ability to effectively filter out all UVB radiations. But the extent of adaptation may vary in different plants, which has been corroborated through research conducted by scientist **Gareth Jenkins[14] at Glasgow University in the UK**. He has been studying the rate of UVB radiation absorption by various types of Brassicaceae plants. In his research, he has demonstrated the effectiveness of "Sinapate Esters" components in blocking sun's harmful UVB radiations. This is why such plants do not show any UVB damage even with prolonged exposure to sunlight. However, as per **Carsten Milkowski**, only Brassicaceae plants[15] can produce the component that inhibits UVB radiation, making them highly resistant and hardy plant variety.

Keeping the conflicting opinions and the importance of the issue in mind, a study was carried out to find answers to the research question, "**How far is the enhanced infiltration of UVA radiations (315-400nm) due to ozone depletion less detrimental than UVB radiations(280-315nm) for the germination and growth of Gerbera (*Gerbera L.*) and Chamomile (*Matricaria chamomilla*) flowering plants?"**

An experiment was conducted for investigating the effects of UVA (315-400nm) and UVB (280-315nm) radiations on the germination and growth of 15 seeds each of **Gerbera (*Gerbera L.)*** and **Chamomile (*Matricaria chamomilla*)** flowering plants over a period of 3 months and comparison was also made with normal light conditions (Controlled Group) simulated through LED lights. Growth parameters, i.e., height and day of bud formation were checked intermittently on Days 30,60 and 90. The impacts of different UV radiations were compared through percentage change in growth parameters of both the plants. The results were further used to analyze the correlation(R^2) between dependent and independent variables and calculated the t-test to authenticate the relevance of the topic chosen. The results of the investigation were intended to provide a definitive conclusion on impact of depletion of ozone on plant productivity. Additionally, since **Chamomile** (***Matricaria chamomilla***) is a Brassicaceae plant and **Gerbera (*Gerbera L.*)** is not, so comparative evaluation of being resistive to UVB radiation exposure would be a useful and beneficial exploration, economically.

---

[14] Jenkins, Gareth I. "The UV-B Photoreceptor UVR8: From Structure To Physiology". The Plant Cell, vol 26, no. 1, 2014, pp. 21-37. Oxford University Press (OUP), doi:10.1105/tpc.113.119446. Accessed 3 Jun 2022.

[15] Neugart, Susanne et al. "The Intrinsic Quality Of Brassicaceous Vegetables: How Secondary Plant Metabolites Are Affected By Genetic, Environmental, And Agronomic Factors". Scientia Horticulturae, vol 233, 2018, pp. 460-478. Elsevier BV, doi:10.1016/j.scienta.2017.12.038. Accessed 22 Jun 2022.

What makes the environmental issue worthy of investigation is its relation to the core field of contemporary research on adverse impacts of human actions which cause ozone depletion and also because the findings will provide a detailed analysis of the topic in consideration. The concluded results will assist in formulating regulatory laws at both local and national level for curbing the anthropogenic causes of ozone depletion. Additionally, at state level, such policies can be framed which are aligned to economic growth plans, like choosing those ornamental flowers to be farmed at commercial scale which are more resistive to UVB radiations. This is especially important during the summer blooms as that is when the state is hit by strong heatwaves which have the presence of more UVB radiations.[16]

2. Background information:

The environmental issue in research question has a great relevance as according to a scientific journal published in **Down to Earth**[17], ozone layer depletion is a major cause of concern among the Indian scientists. Adverse global consequences of ozone layer depletion not only on the ecosystem but also on human race has kept the Indian scientists on high alert to notice any such deviation in the Indian skies as well. As in the words of the head of **National Ozone Centre in New Delhi, S K Srivastava**[18], though our skies do not show any signs of depletion which has also been validated by the research scientists of **Indian Meteorological Department, Mr.** V **Thapyal**[19] and **S M Kulshresta** but the threat remains imminent due to the rapidly increasing production and consumption of industrial goods, such as air conditioners and refrigerators.

The cause of concern could also be attributed to the contradictory observations made by the **director of Development Alternatives**, **Dr. K Chatterji.** Through his calculations, he has demonstrated that the ozone thickness in the upper stratospheric layers over Pune and New Delhi showed a declining trend from October 1980

---

[16] *"Heat Waves And The Importance Of Sun Protection | Mcleish Orlando LLP". Mcleish Orlando LLP, 2018, https://www.mcleishorlando.com/insights/heat-waves-and-the-importance-of-sun-protection/. Accessed 18 Feb 2023.*

[17] *"No Ozone Depletion Over India Yet". Downtoearth.Org.In, 2020, https://www.downtoearth.org.in/news/no-ozone-depletion-over-india-yet-29781#:~:text=Ozone%20profiles%20are%20also%20regularly%20recorded%20using%20balloons.&text=He%20warns%20that%20even%20a,affect%20plant%20and%20fish%20production. Accessed 25 Dec 2022.*

[18] *Thangavel, Sivasakthivel & Reddy, K.K.Siva. (2011). Ozone Layer Depletion and Its Effects: A Review. International Journal of Environmental Science and Development. 2. 30-37. 10.7763/IJESD.2011.V2.93. Accessed 3 Jun 2022.*

[19] *" Sarkar, R.P. And Thapliyal, V. (1988) Climatic Change And Variability. Mausam, 39, 127-138. - References - Scientific Research Publishing ". Scirp.Org, 2022, https://www.scirp.org/(S(i43dyn45teexjx455qlt3d2q))/reference/ReferencesPapers.aspx?ReferenceID=1241182. Accessed 3 Jun 2022.*

to 1983. Further concerns have been raised by the academic researcher, **B N Srivastava[20] of the National Physical Laboratory.** According to him, since India already receives very high degrees of UV-B radiations, so even the slightest ozone layer depletion would put us at greater risks. According to the data shared by him, India experiences UV-B radiations of approximately 290nm wavelength, which is similar to what is experienced by Antarctica during enhanced ozone depletion periods.

The issue has been investigated further by academics like **Kim Valenta et al,**[21] who in their scholarly article, **"Ultraviolet…Colour"**, have shared data on the deleterious impacts of UV-B radiation exposure on plant productivity. Reflecting on the valuable inputs of their exploration becomes inevitable as the state government of Rajasthan is in full swing promotion of commercial horticulture. The rich cultural heritage of Rajasthan has made it a popular destination for party gatherings and weddings, leading to higher demand of flowers. Jaipur, for example, has become a hub for flowers, with daily consumption of around 50,000kg. Decorative flowers such as Roses, Lilies, Gerberas, Bird of Paradise, Rajnigandha, Tulips, Orchids, and Anthuriums are in high demand. However, except for Jasmine, Gerbera, and Rose, all other flowers are being imported from surrounding regions. According to data from Rajasthan Horticulture Department, from 2017-18, only 3,596 hectares of land was used for flower cultivation in Rajasthan, thus, government is taking action to promote horticulture. Subsidies have been provided to farmers at various levels to facilitate the transition to intensive flower cultivation to profit from the growing demand of flowers in the international market also. As per the latest development plans shared by **Agriculture Minister Prabhu Lal Saini**[22], the cultivation of ornamental flowers such as Gerbera, Tulip, Chamomile, and Orchid will be the main focus of the government's plans due to their high commercial viability. Thus, this study will assess the feasibility of these proposed plans.

Gerbera and Chamomile flowering plants were chosen as both of them have a relatively shorter germination period and have higher resistivity to changing environmental conditions so will provide a quick comparative

---

[20] Thangavel, Sivasakthivel & Reddy, K.K.Siva. (2011). Ozone Layer Depletion and Its Effects: A Review. International Journal of Environmental Science and Development. 2. 30-37. 10.7763/IJESD.2011.V2.93. Accessed 3 Jun 2022.

[21] Valenta, Kim et al. "Ultraviolet Radiation Changes Plant Color". BMC Plant Biology, vol 20, no. 1, 2022. Springer Science And Business Media LLC, doi:10.1186/s12870-020-02471-8. Accessed 2 Jun 2022.

[22] 2022, Budget et al. "Rajasthan To Focus On Ornamental Flowers". @Businessline, 2022, https://www.thehindubusinessline.com/news/national/Rajasthan-to-focus-on-ornamental-flowers/article20711067.ece. Accessed 22 Jun 2022.

analysis successfully.[23] Moreover, the new varieties of Chamomile plants are more drought resistant and have a higher essential oil content unlike their predecessors.[24] Both the flowering plants have gained high economic importance as Gerbera has an easy and wide cultural acceptance and is also preferred for religious and festive celebrations apart from being just a decorative curios[25] and the essential oils extracted from chamomile flowers are used for herbal remedies. The plants are also used to produce the world-famous chamomile tea which has many therapeutic benefits.[26]

3. Hypothesis:

3.1 Null Hypothesis:

UVA rays will have a similar negative effect on the productivity of both Gerbera and Chamomile flowering plants as UVB rays.

3.2 Alternate Hypothesis:

The enhanced infiltration of UVA radiations due to ozone depletion will be less detrimental than UVB radiations for the germination and growth of Gerbera (*Gerbera L.*) and Chamomile (*Matricaria chamomilla*) flowering plants as UVA radiations have longer wavelength falling within the range of 315-400nm, unlike the UVB radiations which have a shorter wavelength of 280-315nm. UVB radiations have the capacity of penetrating the cell walls, damaging the morphological structure of the plants, and altering the DNA structure which hampers the physiochemical reactions in the plant system.[27] Moreover, Gerbera plants will show more deterioration as compared to Chamomile plants because the latter belong to brassicaceae family which have the capacity to produce sinapate esters which can cut out sun's UVB radiations.[28]

---

[23] *Bhattarai, Krishna et al. "Sequencing And Analysis Of Gerbera Daisy Leaf Transcriptomes Reveal Disease Resistance And Susceptibility Genes Differentially Expressed And Associated With Powdery Mildew Resistance". BMC Plant Biology, vol 20, no. 1, 2020. Springer Science And Business Media LLC, doi:10.1186/s12870-020-02742-4. Accessed 13 Sept 2022.*

[24] *Rathore, Shalika, and Rakesh Kumar. "Agronomic Interventions Affect The Growth, Yield, And Essential Oil Composition Of German Chamomile (Matricaria Chamomilla L.) In The Western Himalaya". Industrial Crops And Products, vol 171, 2021, p. 113873. Elsevier BV, doi:10.1016/j.indcrop.2021.113873. Accessed 18 Dec 2022.*

[25] *"Gerbera Cultivation | Gerbera Farming | Crop Guide | Gerbera ". Indiaagronet.Com, 2022, https://www.indiaagronet.com/indiaagronet/crop%20info/Gerbera.htm. Accessed 27 Dec 2022.*

[26] *Gupta. "Chamomile: A Herbal Medicine Of The Past With A Bright Future (Review)". Molecular Medicine Reports, vol 3, no. 6, 2010. Spandidos Publications, doi:10.3892/mmr.2010.377. Accessed 18 Jul 2022.*

[27] *Agriculturejournals.Cz, 2022, https://www.agriculturejournals.cz/publicFiles/52840.pdf. Accessed 27 Dec 2022.*

[28] *"Zwier Lab - Purdue University". Chem.Purdue.Edu, 2022, http://www.chem.purdue.edu/zwier/?_ga=2.97920279.906543378.1613986145-1359163524.1613986145. Accessed 22 Jun 2022.*

4. Variables:

4.1 Independent Variables:

Radiations simulated with light bulbs of varying wattage – UVA- 20W, UVB- 13W, Normal Light- 36W.

4.2 Dependent Variables:

Germination and growth of Gerbera (*Gerbera L.*) and Chamomile (*Matricaria chamomilla*) flowering plants.

5. Methodology:[29]

Primary investigation was conducted to study the germination and growth of Gerbera (*Gerbera L.*) and Chamomile (*Matricaria chamomilla*) flowering plants, which were exposed to light bulbs of wavelength- 36W for Control group (simulating Normal light conditions), 20W (simulating UVA radiations) and 13W (simulating UVB radiations) for a period of 3 months. Seeds were sown in the month of May as these are frost free days and have temperatures within the range of 24 - 28^0 C through day and night, which is suitable for optimum growth of plants. Lights were turned off at 6pm every day and were switched on at 6am every morning to replicate the natural surroundings.

5.1 Experimental Set-up:

5.1.1 Seed preparation:

Premium quality seeds were taken for the experiment, which were sterilized in a 2.5% sodium hypochlorite medium. The seeds were inserted for 25 minutes and then washed in distilled water.

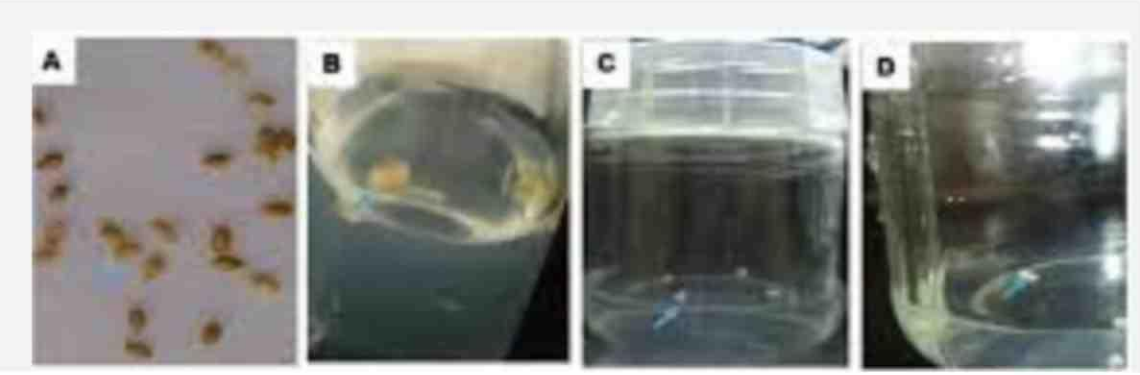


Figure 1: Seed preparation

[29] *Youtube.Com, 2023, https://www.youtube.com/watch?v=riD4vzl34hY. Accessed 27 Aug 2022.*

5.1.2 Seed germination and plant growth set-up:

For germination and further growth, 30 pots of 6inch each were used and were divided into 2 groups of 15 pots each labelled from Pot 1 to Pot 15. Group 1 was for Gerbera plants and Group 2 was for Chamomile plants. 15 pots of both the groups were further divided into 3 Lots with 5 pots each. Lot 1 for LED- Normal light conditions (Control Group), Lot 2 for UV-A radiations; and Lot 3 for UV-B radiations. The pots were then sprayed with Neem oil which is an organic pesticide to avoid any pest attack during the investigation. A mixture of sandy-loam soil and farm manure was prepared in the ratio of 2:1, and 450gm of soil mix was inserted in each pot. Sandy-loam soil was used for the experiment because it is easily friable and is most fertile due to appropriate water holding capacity and good drainage.[30]

1 Gerbera seed was inserted in all the pots of Group 1 and 1 seed of Chamomile was inserted in all the pots of Group 2. It was ensured that all the seeds should be inserted equidistant from each other so as to facilitate appropriate nutrition availability and proper growth of all the seeds. Also, care was taken to insert all the seeds of Gerbera keeping the sharp edge of the seed downward at a depth of 1inch to facilitate optimal growth by providing adequate spacing and sufficient supply of minerals and nutrients and then were covered partially with little soil on the top as light aids in germination but Chamomile seeds were just firmly pressed on to the soil as they need more light to germinate.

Dark room in the school was used for the experiment where pots were left undisturbed for a period of 3months and pots were exposed to their specific & Normal LED, UV-A & UV-B light conditions. Light exposure distinctions were ensured by creating partitions with hard cardboards so that there is no light interference or penetration into the other set up. The set-up included 3 lamps which were fitted with the respective UV radiation bulbs. A gap of 30cm was kept between the lamps and the pots. The room was opened on alternative days to water the seedlings with 100ml water in the initial 10 days which is essential to retain sufficient amount of moisture in the soil which would facilitate plant growth and flowering[31] but after that plants were watered after every 4 days;

---

[30] *Shahid, Shaukat Ali et al. "Improvement In The Water Retention Characteristics Of Sandy Loam Soil Using A Newly Synthesized Poly(Acrylamide-Co-Acrylic Acid)/Alznfe2o4 Superabsorbent Hydrogel Nanocomposite Material". Molecules, vol 17, no. 8, 2012, pp. 9397-9412. MDPI AG, doi:10.3390/molecules17089397. Accessed 18 Feb 2023.*

[31] *"Gerbera". Drysrhu.Edu.In, 2022, http://www.drysrhu.edu.in/crops/Gerbera.html. Accessed 27 Dec 2022.*

and also, to note the germination rate, and to make weekly observations for the growth parameters to procure a comparative analytical data. A water dissolvable super phosphate fertilizer was also applied to the plants after every 20 days as plants would need extra nutrition for growth, but it was ensured that there was no direct fertilizer application on the blooms or leaves as this will lead to rot.

5.2 Sampling strategy & justification:

The systematic sampling method was chosen for the experiment because it is straightforward to implement, analyze, and compare, and eliminates errors that can occur with random sampling. This method has a low risk of corrupting the data.[32]

[32] *"Systematic Sampling: Advantages And Disadvantages". Investopedia, 2022, https://www.investopedia.com/ask/answers/042415/what-are-advantages-and-disadvantages-using-systematic-sampling.asp. Accessed 14 Sept 2022.*

5.3 Photographic Evidence:










6. Data collection & processing:

Table 1: Rate of germination of Gerbera & Chamomile seeds under each light radiation at the End of Week 1		
Light radiation	**Rate of germination of Gerbera seeds**	**Rate of germination of Chamomile seeds**
LED (Lot 1)	Germination of all seeds occurred on the 13th day of sowing	Germination of all seeds occurred on the 13th day of sowing
UV-A (Lot 2)	Germination of all seeds occurred on the 11th day of sowing	Germination of all seeds occurred on the 12th day of sowing
UV-B (Lot 3)	Germination of all seeds occurred on the 9th day of sowing	Germination of all seeds occurred on the 10th day of sowing

Table 2: Height(inch) of Gerbera plants exposed to different light radiations at the End of Day 30								
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average (inch)**	**Percentage change (%) in relation to LED exposure**
LED (Lot 1)	2.50	2.00	1.50	2.50	2.00	**0.42**	**2.10**	**-**
UV-A (Lot 2)	3.00	3.50	3.50	3.00	2.50	**0.42**	**3.10**	**47.62**
UV-B (Lot 3)	4.00	4.50	4.00	4.00	4.50	**0.27**	**4.20**	**100.00**
Height(inch) of Gerbera plants exposed to different light radiations at the End of Day 60								
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average (inch)**	**Percentage change (%) in relation to LED exposure**
LED (Lot 1)	13.00	13.50	14.00	14.00	15.00	**0.74**	**13.90**	**-**
UV-A (Lot 2)	10.50	10.00	9.50	10.50	11.00	**0.57**	**10.30**	**-25.90**
UV-B (Lot 3)	7.50	7.50	7.00	7.50	8.00	**0.35**	**7.50**	**-46.04**
Height(inch) of Gerbera plants exposed to different light radiations at the End of Day 90								
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average (inch)**	**Percentage change (%) in relation to LED exposure**
LED (Lot 1)	35.00	35.50	34.50	35.00	34.50	**0.42**	**34.90**	**-**
UV-A (Lot 2)	22.00	24.00	22.00	21.00	21.50	**1.14**	**22.10**	**-29.72**
UV-B (Lot 3)	13.00	14.00	13.00	12.00	13.00	**0.71**	**13.00**	**-47.79**

Table 3: Average height(in.) of Gerbera plants after Day 30, 60 & 90						
Type of Light radiation	**Day 30**	**SD**	**Day 60**	**SD**	**Day 90**	**SD**
LED (Lot 1)	**2.10**	**0.42**	**13.90**	**0.74**	**34.90**	**0.42**
UV-A (Lot 2)	**3.10**	**0.42**	**10.30**	**0.57**	**22.10**	**1.14**
UV-B (Lot 3)	**4.20**	**0.27**	**7.50**	**0.35**	**13.00**	**0.71**

Graph 1: Representation of the average height(inch) of Gerbera plants after Day 30, 60 & 90

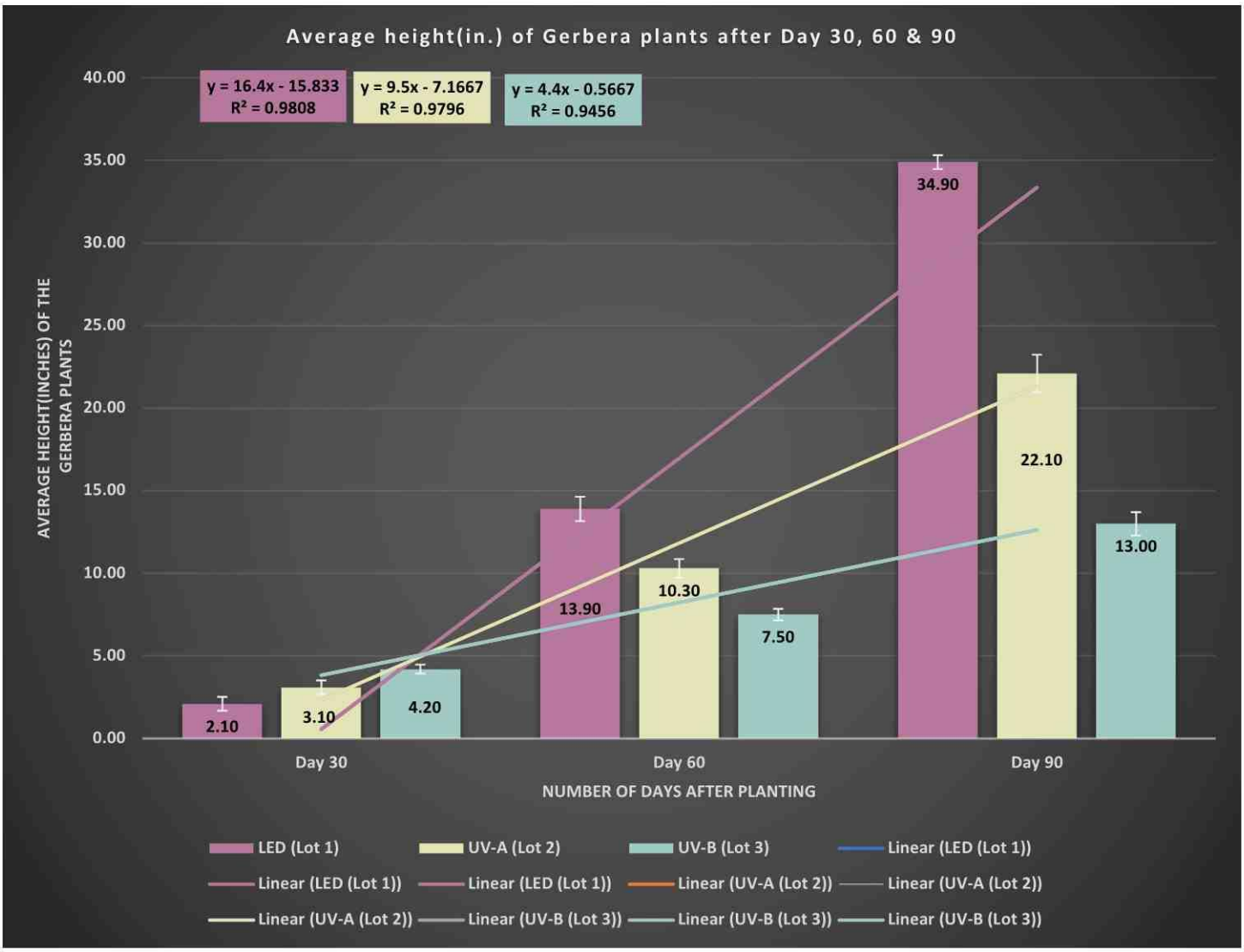


Table 4: Day of bud formation in Gerbera plants on exposure to different light radiations till Day 90							
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average days**
LED (Lot 1)	Day 75	Day 76	Day 75	Day 78	Day 75	**1.30**	**75.80**
UV-A (Lot 2)	Day 89	-	Day 89	Day 88	Day 88	**39.58**	**88.50***
UV-B (Lot 3)	-	-	-	-	-	**-**	**-**

***(For calculating averages only 4 seeds were taken which had formed buds)**

Graph 2: Representation of Day of bud formation in Gerbera plants on exposure to different light radiations till Day 90

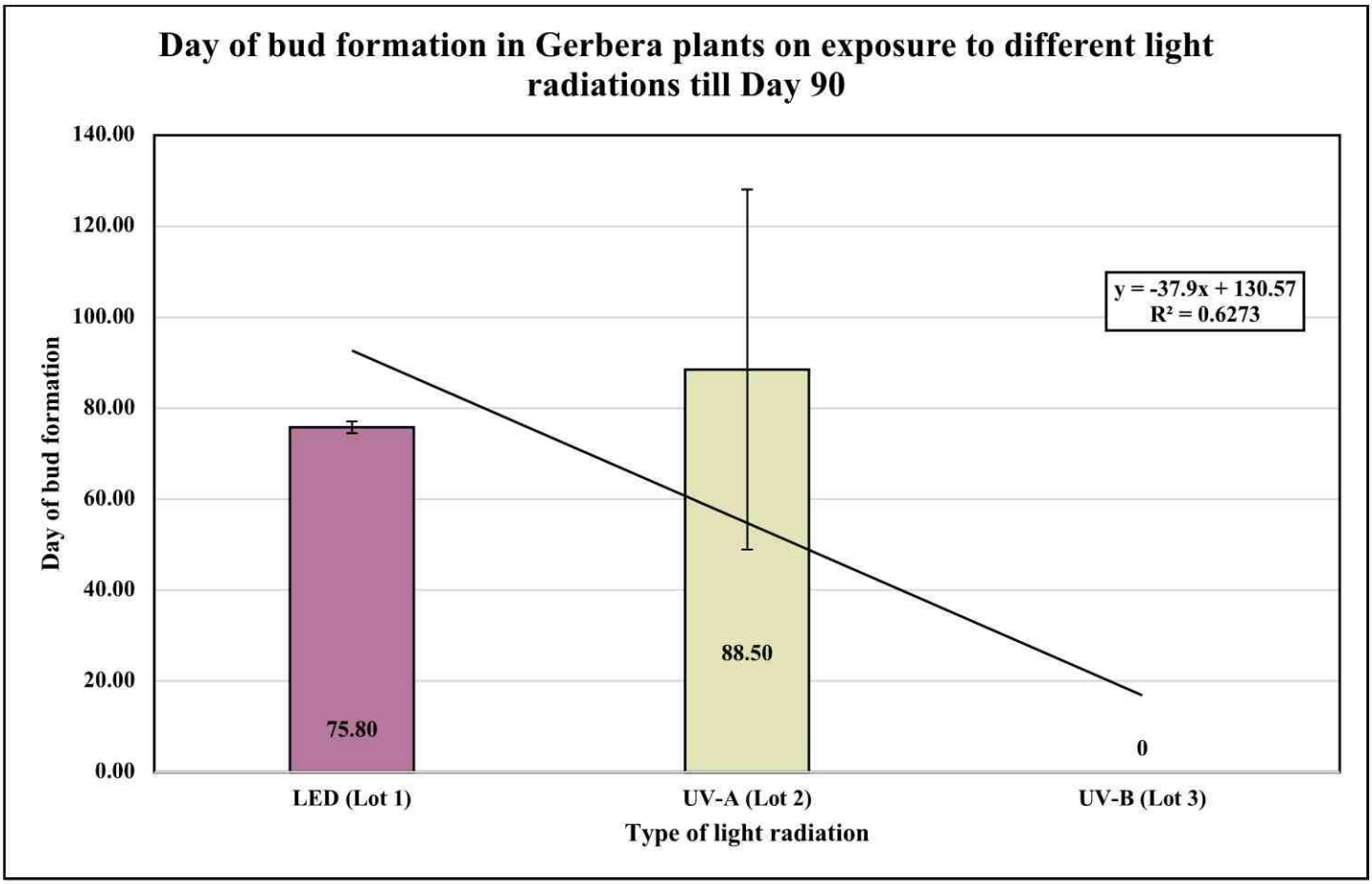


Table 5: Height(inch) of Chamomile plants exposed to different light radiations at the End of Day 30								
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average (inch)**	**Percentage change (%) in relation to LED exposure**
LED (Lot 1)	8.00	8.00	8.00	8.50	7.00	**0.55**	**7.90**	**-**
UV-A (Lot 2)	10.00	10.00	9.50	9.00	9.00	**0.50**	**9.50**	**20.25**
UV-B (Lot 3)	11.50	11.00	10.50	11.00	11.00	**0.35**	**11.00**	**39.24**
Height(inch) of Chamomile plants exposed to different light radiations at the End of Day 60								
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average (inch)**	**Percentage change (%) in relation to LED exposure**
LED (Lot 1)	18.00	18.50	18.00	17.50	17.50	**0.42**	**17.90**	**-**
UV-A (Lot 2)	16.50	16.00	16.00	16.50	15.00	**0.61**	**16.00**	**-10.61**
UV-B (Lot 3)	13.50	13.00	13.00	13.00	14.00	**0.45**	**13.30**	**-25.70**
Height(inch) of Chamomile plants exposed to different light radiations at the End of Day 90								
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average (inch)**	**Percentage change (%) in relation to LED exposure**
LED (Lot 1)	29.00	29.00	29.50	29.00	29.50	**0.27**	**29.20**	**-**
UV-A (Lot 2)	21.50	21.50	20.00	21.00	20.50	**0.65**	**20.90**	**-28.42**
UV-B (Lot 3)	18.50	18.50	18.00	18.00	18.50	**0.27**	**18.30**	**-37.33**

Table 6: Average height(in.) of Chamomile plants after Day 30, 60 & 90						
Type of Light radiation	Day 30	SD	Day 60	SD	Day 90	SD
LED (Lot 1)	7.90	0.55	17.90	0.42	29.20	0.27
UV-A (Lot 2)	9.50	0.50	16.00	0.61	20.90	0.65
UV-B (Lot 3)	11.00	0.35	13.30	0.45	18.30	0.27

Graph 3: Representation of the average height(inch) of Chamomile plants after Day 30, 60 & 90

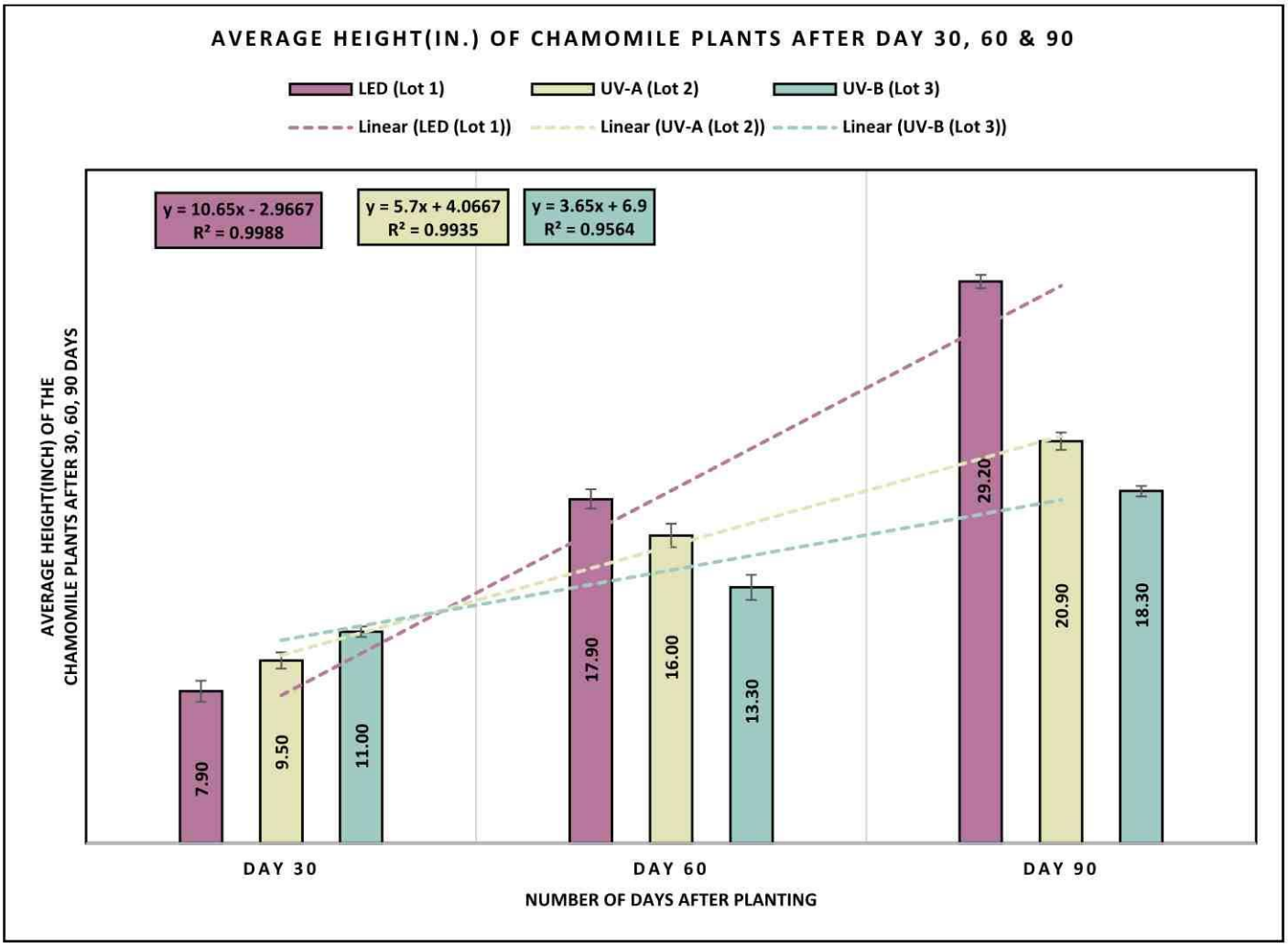


Table 7: Day of bud formation in Chamomile plants on exposure to different light radiations (till Day 90)							
Type of Light radiation	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 4**	**Seed 5**	**Standard Deviation**	**Average**
LED (Lot 1)	Day 60	Day 61	Day 59	Day 58	Day 60	**1.14**	**59.60**
UV-A (Lot 2)	Day 70	Day 71	Day 71	Day 70	Day 72	**0.84**	**70.80**
UV-B (Lot 3)	Day 78	Day 78	Day 81	Day 81	Day 82	**1.87**	**80.00**

Graph 4: Representation of Day of bud formation in Chamomile plants on exposure to different light radiations (till Day 90):

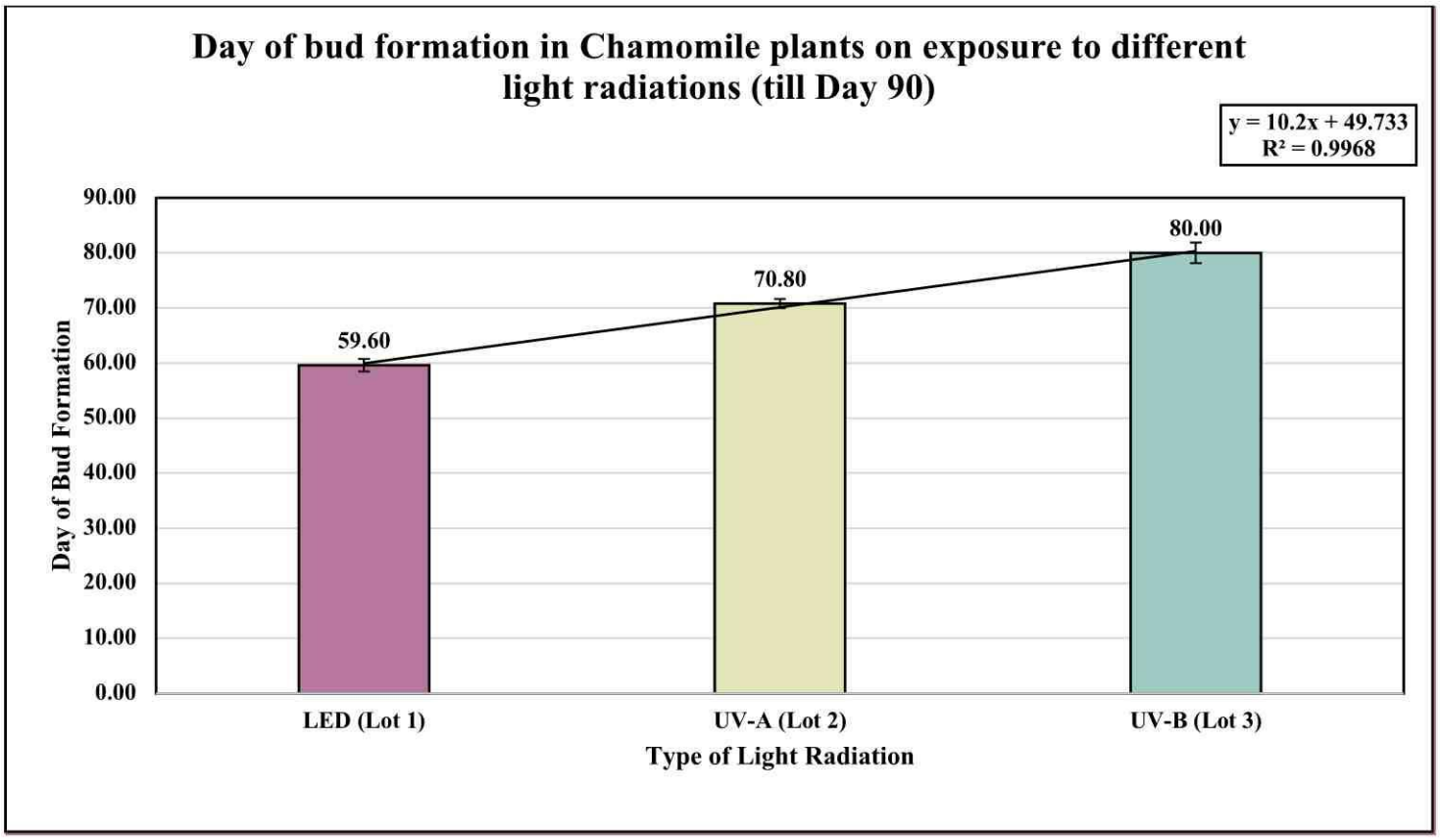


Table 8: Light radiation-wise comparison of the average height(in.) of Gerbera and Chamomile plants after Day 30, 60 & 90:

Type of Light radiation	Day 30		Day 60		Day 90		Difference in height (inch) in relation to UVB Radiations(nm) after 90 days	
	Gerbera	Chamomile	Gerbera	Chamomile	Gerbera	Chamomile	Gerbera	Chamomile
LED (Lot 1)	2.10	7.90	13.90	17.90	34.90	29.20	21.90 (91.54%)	5.50 (59.56%)
UV-A (Lot 2)	3.10	9.50	10.30	16.00	22.10	20.90	9.10 (34.62%)	2.60 (14.21%)
UV-B (Lot 3)	4.20	11.00	7.50	13.30	13.00	18.30	-	-

Graph 5: Representation of comparison in height (in.) of Gerbera and Chamomile plants based on exposure to different light radiations after 30, 60 and 90 days:

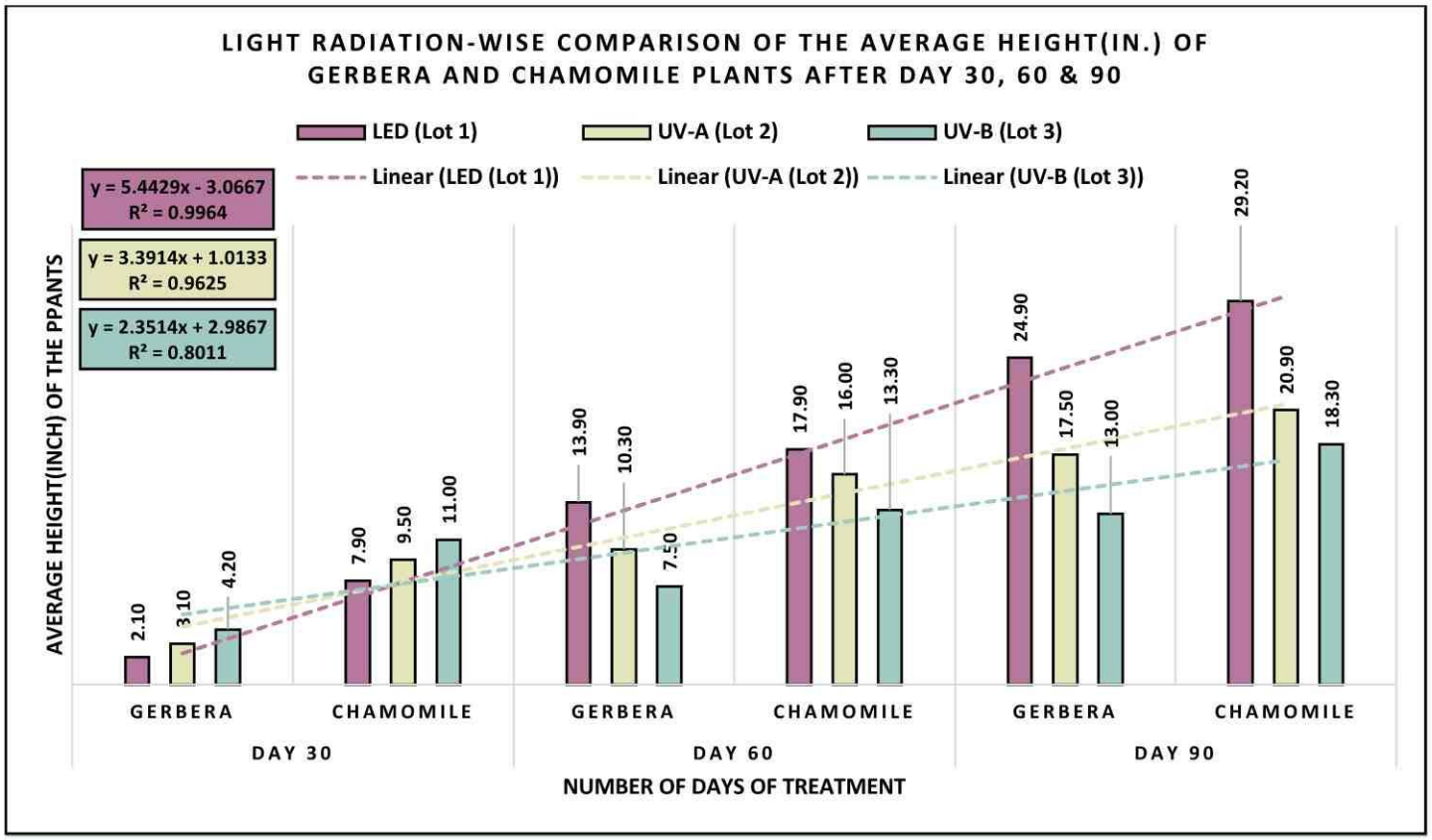
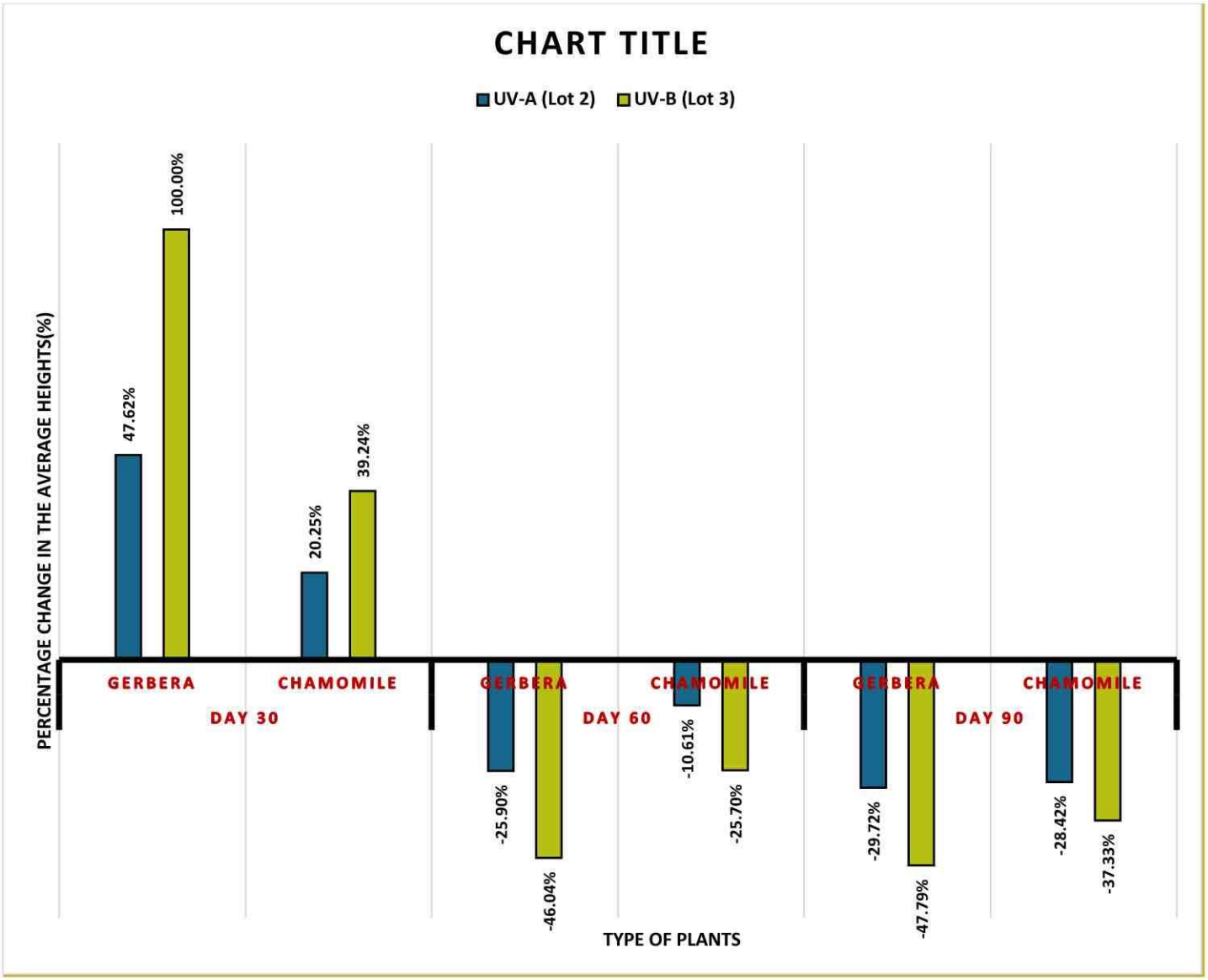


Table 9: A comparison of the average height (in inches) of Gerbera and Chamomile plants under different light radiations, represented as a percentage change (%), at the end of days 30, 60, and 90.

Type of Light radiation	Day 30		Day 60		Day 90	
	Gerbera	Chamomile	Gerbera	Chamomile	Gerbera	Chamomile
LED (Lot 1)	-	-	-	-	-	-
UV-A (Lot 2)	47.62%	20.25%	-25.90%	-10.61%	-29.72%	-28.42%
UV-B (Lot 3)	100.00%	39.24%	-46.04%	-25.70%	-47.79%	-37.33%

Graph 6: Percentage difference in average heights of Gerbera and Chamomile plants with increasing exposure of 30, 60, 90days to UVA & UVB lights with respect to LED lights:



7. Result Analysis & Discussion:

7.1 Qualitative observation:

Weeks	LED Light	UVA Light	UVB Light
After Day 30	Germination of both plants occurred at last Gerbera: First true leaves appeared very heathy green with visible venation Chamomile: True leaves had appeared which seemed a healthy floral green;	Both Gerbera and Chamomile seeds germinated second and the cotyledons had completely opened; but the leaves of both the plants had started curling at the edges although the impact on the chamomile plants was visibly lesser as compared to Gerbera plants	Seeds of both the plants germinated first with unfurled cotyledons; some of the leaves were still in an unfurling situation but in Gerbera plants, even the ones that had opened did not seem to be too healthy
After Day 60	Gerbera: Leaves had proper development of venation; they were pubescent on the under surface and properly lobed. The plants stood straight with erect slender stems. Chamomile: Leaves were long and narrow; some leaves were bi-pinnate and some tri-pinnate; stems appeared strong.	Gerbera: Plants did not look as heathy as their counterparts under LED light exposure; some plants seemed droopy; even number of leaves were less; and had burnt edges Chamomile: Plants were certainly better than Gerbera plants and leaves although not completely healthy but didn't have burnt edges; plants stood straighter although their appearance appeared sickly in comparison.	Gerbera: Burnt edges; brown splotches scattered on all the leaves; stems had started dropping Chamomile: Burnt edges; brown splotches scattered on all the leaves; stems had started dropping
After Day 90	Gerbera: The leaves were all arranged in a rosette pattern, with an elliptical shape and edges were either toothed, or lobed. They were pinnately veined, leathery and had a felt-like texture on the underside; Straight standing stems; foliage was dark green with tuberous stems. Chamomile: leaves continued to grow strong; plants had straight standing stems which were highly ramified; foliage is thick with yellow green leaves and stems.	Gerbera: Plants seemed sick with drooping stems and brittle leaves and burnt edges; it seemed that many will not be able to bear flowers Chamomile: Plants seemed better than Gerbera plants, but their stems were also drooping; the edges of the leaves were curling and some had burnt edges	Gerbera: Some plants seemed completely burnt with stems that were curled which could not stand straight; leaves were powdery to touch as they were very brittle Chamomile: Although these plants also didn't seem to survive for long but their condition was better than Gerbera plants as could also be seen through the quantitative data

7.2 Quantitative observation:

Data presented in **Tables 1 to 9 & plotted in Graphs 1 to 6,** provide a conclusive data for the environmental issue suggested in the research question.

As can be seen from **Table 1** that the seeds of both the plants Gerbera and Chamomile reacted well to UV-A & UV-B radiation exposure, as all the seeds exposed to LED germinated on the 13th day of sowing, whereas, the Gerbera seeds germinated on the 11th day and Chamomile seeds germinated on the 12th day under UV-A exposure; also, the Gerbera seeds germinated on the 9th day and Chamomile seeds germinated on the 10th day under UV-B exposure.

Positive response of plants to higher UV exposure continued till Day 30, but began declining beyond that. **Table 2 & 3; Graph 1** show the average height(in.) of Gerbera plants under exposure to different radiations with increasing number of days. Plants exposed to LED radiations showed an average height of 2.10inch, 13.0inch, and 34.90inch at Day 30, Day 60, Day 90, respectively; whereas plants exposed to UV-A radiations showed average height of 3.10inch, 10.30inch, and 22.10inch, respectively; and plants exposed to UV-B radiations showed average height of 4.20inch, 7.50inch, and 13.00inch, respectively. Similarly, **Table 5 & 6; Graph 3** show the average height(in.) of Chamomile plants. Plants exposed to LED radiations showed an average height of 7.90inch, 17.90inch, and 29.20inch at Day 30, Day 60, Day 90, respectively; whereas plants exposed to UV-A radiations showed average height of 9.50inch, 16.00inch, and 20.90inch, respectively; and plants exposed to UV-B radiations showed average height of 11.00inch, 13.30inch, and 18.30inch, respectively. **(Table 8; Graph 5)**

Graph 1 also corroborates the result through the values of correlation and y equation between type of plant gerbera and Chamomile to increasing exposure to type of radiation – LED, UV-A, &UV-B, respectively. Gerbera plants show strong correlation values of R^2= 0.9808 and high y equation value= 16.4x – 15.833 in LED exposure; R^2 values decrease to 0.9796; y= 9.5x – 7.1667 in UV-A exposure; and falls further to R^2= 0.9456 and y= 4.4x – 0.5667 in exposure to UV-B radiations. Similarly, Chamomile plants show strong correlation values of R^2= 0.9988 and high y equation value= 10.65x – 2.9667 in LED exposure; R^2 values decrease to 0.9935; y= 5.7x + 4.0667 in UV-A exposure; and falls further to R^2= 0.9564 and y= 3.65x + 6.9 in exposure to UV-B radiations (**Graph 3**).

Further, **Table 9; Graph 6,** show the plant-wise comparison of percentage change (%) in average heights(inch) of both the plants in each of the light radiations at the end of Day 30, 60 & 90, with respect to LED light exposure. It was observed that with UV-A exposure, height of Gerbera plants increased by 47.62% and Chamomile plants increased by 20.25% till Day 30; then Gerbera plant height declined to 25.90% but only by 10.61% in Chamomile plants at Day 60; further, then Gerbera plant height declined to 29.72% and by 28.42% in Chamomile plants at Day 90. The major decline was recorded in both the plants which were exposed to UV-B radiations as, height of Gerbera plants increased by 100.00% and Chamomile plants increased by 39.24% till Day 30; then Gerbera plant height declined by 46.06% but only by 25.70% in Chamomile plants at Day 60; further, then Gerbera plant height declined to 47.79% and by 37.77% in Chamomile plants at Day 90.

Table 4; Graph 2, show the day of bud formation of Gerbera plants. Plants exposed to LED radiations showed a bud formation on an average day of 75.80 as opposed to the Chamomile seeds which showed bud formation on average 59.60days; Gerbera plants exposed to UV-A radiations showed a bud formation on an average day of 88.50 as opposed to the Chamomile seeds which showed bud formation on average 70.80days; and Gerbera plants exposed to UV-B radiations showed no bud formation even till the end of the experiment, i.e., Day 90 against the Chamomile seeds which showed bud formation on average 80days **(Table 7; Graph 4).**

The weak R^2 values of 0.6273 and negative y equation values= -37.9x + 130.57 **(Graph 2)** and strong R^2 values of 0.9968 and positive y equation values = 10.2x + 49.733 **(Graph 4)** for Chamomile plants shows the resistivity of these plants against the Gerbera plants.

Lower standard deviation values authenticate the validity of the methodology adopted and the reliability of the data collected.

It can be determined that adverse impacts of climate change on plant growth vary in different global regions. However, due to regional climate differences and availability of natural resources, it is challenging to predict crop responses at local levels by using global models. UV exposure sensitivity of plants can be determined by damage to cell components or ineffective acclimatization processes.[33] UV-B radiation exposure cause leaf differentiation

[33] *"Specific Effects Of UVB Radiation On Plants - Ultraviolet Radiation". Climate Policy Watcher, 2023, https://www.climate-policy-watcher.org/ultraviolet-radiation-2/specific-effects-of-uvb-radiation-on-plants.html. Accessed 18 Feb 2023.*

and senescence by modifying leaf structure. According to scientists like **Greenberg et al.,**[34] UV-B radiation exposure reduces heights of plants, area of leaves, and dry weight, while also increasing auxiliary branches and curling of leaves. They have found that on UV-B exposure for few weeks, the area of leaves and dry weight of rice was reduced significantly. Increased exposure to UV-A radiation can also harm plant growth as it decreases the leaf area per unit plant biomass. High levels of UV-B have been shown to significantly decrease relative rate of growth of leaf area, its ratio and nitrogen productivity.

The decline in crop production due to ultraviolet radiation is primarily caused by damaged organ membranes, like of chloroplasts, which in turn leads to indirect effects on plant growth from other stressors like oxidative stress. Increased UV-B radiation from sun reduces photosynthesis and crop production in some of the plant species. While plants are generally equipped to withstand UV exposure under normal growth conditions, excessive exposure can lead to oxygen toxicity when protective systems are overwhelmed by environmental stressors like drought, intensive light, higher salinity levels, water logging, air pollution, and toxicity by heavy metals. Studies have indicated that even ambient levels of UV-B radiation can trigger ecosystem stress as it damages plant cells, proteins, membrane lipids, and DNA.[35]

According to **science professors Chen Shi et al**,[36] plants can adapt to certain levels of UV-B radiations and light intensity, but the tolerance range varies based on the species, age, duration of exposure, etc. When UV-B exposure exceeds the tolerance limits, the anatomy of the plant leaves changes and biomass decreases. UV-B radiation also elicits protective response among plants, like alterations in antioxidant enzyme activity and PA content. Some plant species, like Chamomile, which belong to the Brassicaceae family, show better tolerance to UV radiation exposure due to their production of secondary metabolites, such as flavonoids and anthocyanins, which absorb UV radiations, and stop them from penetrating the mesophyll cells present in leaves.[37] There is complex

---

[34] Greenberg BM, Wilson MI, Huang XD, Duxbury CL, Gerhaddt KE, Gensemer RW (1997). The effects of ultraviolet-B radiation on higher plants. In: Wang W., Goursuch J., Hughes J.S. (eds.): Plants for environmental studies. Boca Raton, Fl: CRC Press, pp. 1–35. Accessed 7 Nov 2022.

[35] Nawkar, Ganesh et al. "UV-Induced Cell Death In Plants". International Journal Of Molecular Sciences, vol 14, no. 1, 2013, pp. 1608-1628. MDPI AG, doi:10.3390/ijms14011608. Accessed 18 Feb 2023.

[36] Shi, Chen, and Hongtao Liu. "How Plants Protect Themselves From Ultraviolet-B Radiation Stress". Plant Physiology, vol 187, no. 3, 2021, pp. 1096-1103. Oxford University Press (OUP), doi:10.1093/plphys/kiab245. Accessed 8 Feb 2023.

[37] Marciniak, Pawel et al. "Differentiated Effects Of Secondary Metabolites From Solanaceae And Brassicaceae Plant Families On The Heartbeat Of Tenebrio Molitor Pupae". Toxins, vol 11, no. 5, 2019, p. 287. MDPI AG, doi:10.3390/toxins11050287. Accessed 7 Nov 2022.

antioxidant system present in plants that protects them from oxidative damage,[38] but the other plant species may react deleteriously to UV-B exposures.

8. Conclusion:

Results presented in **Tables 1 to 9; Graphs 1 to 6**, validate the alternate hypothesis made at beginning of the experiment, that the enhanced infiltration of UVA radiations due to ozone depletion will be less detrimental than UVB radiations for the germination and growth of Gerbera (*Gerbera L.*) and Chamomile (*Matricaria chamomilla*) flowering plants. In addition, it was also assumed that Gerbera plants will show more deterioration as compared to Chamomile plants because the latter belong to brassicaceae family also stands correct.

Table 1, shows that UV-B radiations promoted the rate of germination in both Gerbera and Chamomile seeds as all seeds germinated on 13thday in LED exposure; all Gerbera seeds germinated on 11thday and all Chamomile seeds germinated on 12thday under UV-A radiations; and all Gerbera seeds germinated on 9thday and all Chamomile seeds germinated on 10thday under UV-B radiations.

Further, inferences can be made (**Table 9; Graph 6),** initially after Day 30, Gerbera plants showed a higher percentage increase of 47.62% and 100.00% in UV-A & UV-B radiation exposure, whereas Chamomile plants showed a percentage increase of only 20.25% and 39.24%, respectively; but trends took a complete 360^0 turn at Day 60 where Gerbera plants showed a percentage decline of 25.09% and 46.04% in UV-A & UV-B radiation exposure, whereas Chamomile plants showed a percentage decline of only 10.61% and 25.70%, respectively. Further, at Day 90, percentage decline in height of Gerbera plants is 29.72% in UV-A and 47.79% in UV-B exposure as compared to 28.42% in UV-A and 37.77% in UV-B exposure in Chamomile plants. Although, there is a difference of only 1.30%, yet the decline is more in Gerbera plants, which implies that the sensitivity of Gerbera plants towards UV radiation exposure is more as compared to Chamomile plants.

Further validations come through correlation between type of radiation and plant height (**Graph 1 & 3),** there is very strong R^2 value between LED exposure and Gerbera and Chamomile plant, respectively– 0.9808 and 0.9988;

---

[38] Racchi, Milvia. "Antioxidant Defenses In Plants With Attention To Prunus And Citrus Spp.". Antioxidants, vol 2, no. 4, 2013, pp. 340-369. MDPI AG, doi:10.3390/antiox2040340. Accessed 18 Feb 2023.

a little weaker correlation of R^2 0.9796 and 0.9935 with UV-A exposure and the weakest correlation, R^2 of 0.9456 and 0.9564 with UV-B radiation exposure.

The correlation value for the bud formation in Gerbera plants was a weak R^2 value of 0.6273 and a negative y value of = -37.9x + 130.57; whereas, Chamomile plants showed a strong R^2 value of 0.9968 and a positive y value of = 10.2x + 49.733.

Thus, it can be safely assumed that UV-B radiations of 315-400nm wavelength are much more harmful than the UV-A radiations of 280-315nm wavelength and the normal sunlight which is represented by the LED lights.

9. Limitation and possible improvements:

The study had some strengths, such as using five trial plants for each exposure to light, providing a sufficient sample size to obtain reliable data. However, some limitations were also present, such as:

- 90-day duration of the experiment only allowing for limited investigation into the advanced growth parameters of the plants. Further exploration would reveal more information about the impacts of LED, UVA, and UVB radiation on plant morphology such as foliage cover, number of flowers, and flower width.
- Additionally, the experiment was conducted in a closed dark chamber, which did not accurately reflect natural environmental conditions, such as wind and humidity. In the future, similar experiments could be conducted in boxes with holes for proper air circulation to account for these factors.

10. Future scope:

As per the results obtained UV-B radiations are far more harmful than the UV-A radiations for both Gerbera and Chamomile plant productivity but the adverse impacts are more apparent in Gerbera plants as compared to the Chamomiles as the latter are Brassicaceae plants which have better adaptability to exposure to UV-B radiations. Thus, further experimentation can be done on more varieties of plants which have better adaptations to the changing climatic conditions, like Petunia and Nigella flowers; and Eggplants and Tomato vegetable plants for result validation and universal applicability of the findings. In addition to that, investigating impacts on other morphological features of the plant also like root length, stem length, number of leaves and flowers, root, shoot and total dry weight would also provide a stronger support to the hypothesis.

11. Bibliography:

- " Sarkar, R.P. And Thapliyal, V. (1988) Climatic Change And Variability. Mausam, 39, 127-138. - References - Scientific Research Publishing ". Scirp.Org, 2022, https://www.scirp.org/(S(i43dyn45teexjx455qlt3d2q))/reference/ReferencesPapers.aspx?ReferenceID=1241182. Accessed 3 Jun 2022.
- "Gerbera Cultivation | Gerbera Farming | Crop Guide | Gerbera ". Indiaagronet.Com, 2022, https://www.indiaagronet.com/indiaagronet/crop%20info/Gerbera.htm. Accessed 27 Dec 2022.
- "Gerbera". Drysrhu.Edu.In, 2022, http://www.drysrhu.edu.in/crops/Gerbera.html. Accessed 27 Dec 2022.
- "Heat Waves And The Importance Of Sun Protection | Mcleish Orlando LLP". Mcleish Orlando LLP, 2018, https://www.mcleishorlando.com/insights/heat-waves-and-the-importance-of-sun-protection/. Accessed 18 Feb 2023.
- "No Ozone Depletion Over India Yet". Downtoearth.Org.In, 2020, https://www.downtoearth.org.in/news/no-ozone-depletion-over-india-yet-29781#:~:text=Ozone%20profiles%20are%20also%20regularly%20recorded%20using%20balloons.&text=He%20warns%20that%20even%20a,affect%20plant%20and%20fish%20production. Accessed 25 Dec 2022.
- "Potential Impacts Of Increased Solar UV-B On Global Plant Productivity". Ciesin.Columbia.Edu, 2022, http://www.ciesin.columbia.edu/docs/011-467/011-467.html. Accessed 22 Aug 2022.
- "Specific Effects Of UVB Radiation On Plants - Ultraviolet Radiation". Climate Policy Watcher, 2020, https://www.climate-policy-watcher.org/ultraviolet-radiation-2/specific-effects-of-uvb-radiation-on-plants.html. Accessed 25 Dec 2022.
- "Study Gets To The Roots Of Plants' Natural Sunscreen - Purdue University". Purdue.Edu, 2022, https://www.purdue.edu/newsroom/releases/2014/Q4/study-gets-to-the-roots-of-plants-natural-sunscreen.html. Accessed 22 Jun 2022.
- "Systematic Sampling: Advantages And Disadvantages". Investopedia, 2022, https://www.investopedia.com/ask/answers/042415/what-are-advantages-and-disadvantages-using-systematic-sampling.asp. Accessed 14 Sept 2022.
- "Ultraviolet (UV) Radiation | Center For Science Education". Scied.Ucar.Edu, 2022, https://scied.ucar.edu/learning-zone/atmosphere/ultraviolet-uv-radiation. Accessed 23 Aug 2022.

- "Zwier Lab - Purdue University". Chem.Purdue.Edu, 2022, http://www.chem.purdue.edu/zwier/?_ga=2.97920279.906543378.1613986145-1359163524.1613986145. Accessed 22 Jun 2022.
- 2022, Budget et al. "Rajasthan To Focus On Ornamental Flowers". @Businessline, 2022, https://www.thehindubusinessline.com/news/national/Rajasthan-to-focus-on-ornamental-flowers/article20711067.ece. Accessed 22 Jun 2022.
- Agriculturejournals.Cz, 2022, https://www.agriculturejournals.cz/publicFiles/52840.pdf. Accessed 27 Dec 2022.
- Bhattarai, Krishna et al. "Sequencing And Analysis Of Gerbera Daisy Leaf Transcriptomes Reveal Disease Resistance And Susceptibility Genes Differentially Expressed And Associated With Powdery Mildew Resistance". BMC Plant Biology, vol 20, no. 1, 2020. Springer Science And Business Media LLC, doi:10.1186/s12870-020-02742-4. Accessed 13 Sept 2022.
- Department of the Environment and Energy. “Department of Agriculture, Water and the Environment.” Department of Agriculture, Water and the Environment, 2018, www.environment.gov.au/protection/ozone/publications/ozone-depleting-substances-factsheet#:~:text=Ozone%20depleting%20substances%20are%20chemicals. Accessed 25 Nov. 2022.
- Fiscus, Edwin L., and Fitzgerald L. Booker. “Is Increased UV-B a Threat to Crop Photosynthesis and Productivity?” Photosynthesis Research, vol. 43, no. 2, Feb. 1995, pp. 81–92, doi:10.1007/bf00042965. Accessed 1 Dec. 2022.
- Greenberg BM, Wilson MI, Huang XD, Duxbury CL, Gerhaddt KE, Gensemer RW (1997). The effects of ultraviolet-B radiation on higher plants. In: Wang W., Goursuch J., Hughes J.S. (eds.): Plants for environmental studies. Boca Raton, Fl: CRC Press, pp. 1–35. Accessed 7 Nov 2022.
- Gupta. "Chamomile: A Herbal Medicine Of The Past With A Bright Future (Review)". Molecular Medicine Reports, vol 3, no. 6, 2010. Spandidos Publications, doi:10.3892/mmr.2010.377. Accessed 18 Jul 2022.
- Jenkins, Gareth I. "The UV-B Photoreceptor UVR8: From Structure To Physiology". The Plant Cell, vol 26, no. 1, 2014, pp. 21-37. Oxford University Press (OUP), doi:10.1105/tpc.113.119446. Accessed 3 Jun 2022.
- Marciniak, Paweł et al. "Differentiated Effects Of Secondary Metabolites From Solanaceae And Brassicaceae Plant Families On The Heartbeat Of Tenebrio Molitor Pupae". Toxins, vol 11, no. 5, 2019, p. 287. MDPI AG, doi:10.3390/toxins11050287. Accessed 7 Nov 2022.

- Nawkar, Ganesh et al. "UV-Induced Cell Death In Plants". International Journal Of Molecular Sciences, vol 14, no. 1, 2013, pp. 1608-1628. MDPI AG, doi:10.3390/ijms14011608. Accessed 18 Feb 2023.
- Neugart, Susanne et al. "The Intrinsic Quality Of Brassicaceous Vegetables: How Secondary Plant Metabolites Are Affected By Genetic, Environmental, And Agronomic Factors". Scientia Horticulturae, vol 233, 2018, pp. 460-478. Elsevier BV, doi:10.1016/j.scienta.2017.12.038. Accessed 22 Jun 2022.
- Nguyen, V. P. Thinh et al. "Sinapic Acid And Sinapate Esters In Brassica: Innate Accumulation, Biosynthesis, Accessibility Via Chemical Synthesis Or Recovery From Biomass, And Biological Activities". Frontiers In Chemistry, vol 9, 2021. Frontiers Media SA, doi:10.3389/fchem.2021.664602. Accessed 8 Feb 2023.
- Potdar, Aditee. "Business Initiatives For Conservation And Enhancement Of Biodiversity". Journal Of Ecosystem & Ecography, vol 01, no. S1, 2012. OMICS Publishing Group, doi:10.4172/2157-7625.s1.009. Accessed 25 Dec 2022.
- Racchi, Milvia. "Antioxidant Defenses In Plants With Attention To Prunus And Citrus Spp.". Antioxidants, vol 2, no. 4, 2013, pp. 340-369. MDPI AG, doi:10.3390/antiox2040340. Accessed 18 Feb 2023.
- Rathore, Shalika, and Rakesh Kumar. "Agronomic Interventions Affect The Growth, Yield, And Essential Oil Composition Of German Chamomile (Matricaria Chamomilla L.) In The Western Himalaya". Industrial Crops And Products, vol 171, 2021, p. 113873. Elsevier BV, doi:10.1016/j.indcrop.2021.113873. Accessed 18 Dec 2022.
- Shahid, Shaukat Ali et al. "Improvement In The Water Retention Characteristics Of Sandy Loam Soil Using A Newly Synthesized Poly(Acrylamide-Co-Acrylic Acid)/Alznfe2o4 Superabsorbent Hydrogel Nanocomposite Material". Molecules, vol 17, no. 8, 2012, pp. 9397-9412. MDPI AG, doi:10.3390/molecules17089397. Accessed 18 Feb 2023.
- Shi, Chen, and Hongtao Liu. "How Plants Protect Themselves From Ultraviolet-B Radiation Stress". Plant Physiology, vol 187, no. 3, 2021, pp. 1096-1103. Oxford University Press (OUP), doi:10.1093/plphys/kiab245. Accessed 8 Feb 2023.
- Thangavel, Sivasakthivel & Reddy, K.K.Siva. (2011). Ozone Layer Depletion and Its Effects: A Review. International Journal of Environmental Science and Development. 2. 30-37. 10.7763/IJESD.2011.V2.93. Accessed 3 Jun 2022.

- UV?, What et al. "Radiation: Ultraviolet (UV) Radiation". Who.Int, 2022, https://www.who.int/news-room/questions-and-answers/item/radiation-ultraviolet-(uv). Accessed 23 Aug 2022.
- Valenta, Kim et al. "Ultraviolet Radiation Changes Plant Color". BMC Plant Biology, vol 20, no. 1, 2022. Springer Science And Business Media LLC, doi:10.1186/s12870-020-02471-8. Accessed 2 Jun 2022.
- Vanhaelewyn, Lucas et al. "Ultraviolet Radiation From A Plant Perspective: The Plant-Microorganism Context". Frontiers In Plant Science, vol 11, 2020. Frontiers Media SA, doi:10.3389/fpls.2020.597642. Accessed 23 Aug 2022.
- Youtube.Com, 2023, https://www.youtube.com/watch?v=riD4vzl34hY. Accessed 27 Aug 2022.